

Afeef Kallanthodan

Systems Engineer | Model Based Design | Embedded SW Development and Testing

14/04/1992

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Indian

EU Blue Card

About Me

Engineer with 8+ years' experience spanning aerospace system integration (5th-gen fighter programme, multi-disciplinary coordination) and automotive embedded SW development. Solution-oriented, structured working style. B1 certified in German.

Languages

English ●●●●●

German(B1) ●●●○○

Skills

- Embedded C / AUTOSAR / MISRA
- MBD - Matlab/Simulink/m-Scripting
- SW Testing - MUT/MIL/SIL/HIL
- SW Requirements (DOORS/Polarion)
- Python / Test Automation

Hobbies

- Real Simulation Games
- Read/watch latest technologies
- AI Enthusiast / Implementing local AI

Working Experience

Sep'24-Now — **Embedded SW Development/Test Engineer**
Koosys GmbH | Regensburg — 2 year 0 months

• Developed MBD-based FOC algorithms for 6-phase IPMSM; generated AUTOSAR ASIL-B compliant code via Embedded Coder.
• Built 6-phase IPMSM Plant Model and MIL environment for performance validation; tuned control parameters on MIL and real HW.
• Created integration test environment (CANoe/WinIDEA/C#), Rest Bus Simulation, and L/R parameter estimation on real HW (SWE5/SWE6).
• Automated unit test generation (Python), CI-CD pipeline setup, and XCP-based SW debugging.
Tools: MATLAB/Simulink, Embedded Coder, XCP, CANoe, WinIdea, CppUTest, Helix QAC, Polarion, Git, Python, C/C#

Jul'23-Aug'24 — **Embedded Software/Test Engineer**
Valentum Engineering GmbH | Regensburg — 1 year 1 month
External Employee at Koosys GmbH | Regensberg

• Conducted unit, integration, and system testing (SWE4–SWE6).
• Performed static code analysis with Helix QAC and automated testing with Java/CAPL.
• Debugged hardware (WinIDEA) and simulated networks with CANoe.
Tools: Eclipse, Java, TESSY, WinIdea, CANoe 17, Git, Helix QAC, CppUTest, DOORS

Jul'21-Mar'23 — **Model Based Design Engineer**
TATA Elxsi Ltd. | Bangalore — 1 year 8 months

• Developed AUTOSAR-based EPS components (ASIL-D) with MBD (Simulink/Stateflow) and unit testing.
• Generated C code, performed static model analysis (MXAM/Polyspace), and validated via MIL/SIL.
• Ensured compliance with ASPICE, ISO 26262, MISRA C, MAAB; Agile environment focus.
Tools: MATLAB/Simulink, StateFlow, MXAM, Polyspace, Embedded Coder, TASKING, GHS, Davinci, Reactis, DOORS/DNG, RTC, Synergy, Git, JIRA, Python

Sep'17–Jul'21 — **Aerospace Systems Engineer**
Aeronautical Development Agency (ADA) | Bangalore, India — 3 years 10 months

• Aircraft-level system integration for 5th-gen fighter program; coordinated hydraulic, ECS and electrical subsystems across multi-disciplinary teams.
• Evaluated subsystem architectures; resolved interface conflicts and developed technical solutions for system-level integration problems.
• Managed bidirectional requirements traceability — aircraft-level HLRs to subsystem specifications using DOORS.
• Developed MATLAB/Simulink/Simscape simulation models for subsystem behaviour analysis; automated mass property scripts.
• Mechanical integration studies (weapon bay, landing gear) in Catia V5.
Tools: MATLAB/Simulink, Simscape, Stateflow, Catia V5, Teamcenter (PLM), DOORS
Standards: MIL-STD, **Self-Study**:DO-178C,ARP 4754A,ARP 4761A *[AFuzion, DO-178CDevelopment of Safety-Critical Software - Leanna Rierson]*

Education

University Education

2011 – 2015 — **Bachelors in Mechanical Engineering** — MG University
Focus: Mechatronics and control system, engineering mathematics, programming in C, etc.

Awards

2022 — **Extra Mile Award** — TATA Elxsi Ltd
Awarded by TATA Elxsi Ltd for delivering two complex and critical software components on scheduled time.

Afeef Kallanthodan **October 8, 2026**